

Assembly instructions for: **Electronic_Eyepiece_for_CDK20**

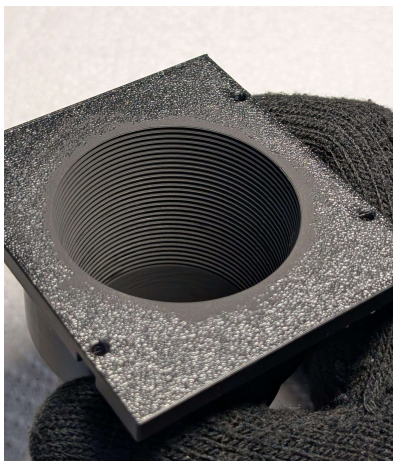
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The electronic eyepiece assembly comprises an astronomical camera, extension tubes, filter changer, optical reducer (which also functions as a 2-inch insertion barrel), display module, and keypad.

The camera, extension tubes, optical reducer, and filter changer are integrated into a permanently assembled optical unit. The outer housing, which incorporates the display and keypad, is mounted on a rotatable support and can be rotated around the camera body to provide a convenient viewing and operating position.

Paint relevant surfaces near display with black matte paint:



Insert first lens, pay attention to the lens orientation:



Insert spacer ring:



Insert grey ND filter:



Insert second lens spacer-holder, pay attention to the orientation:



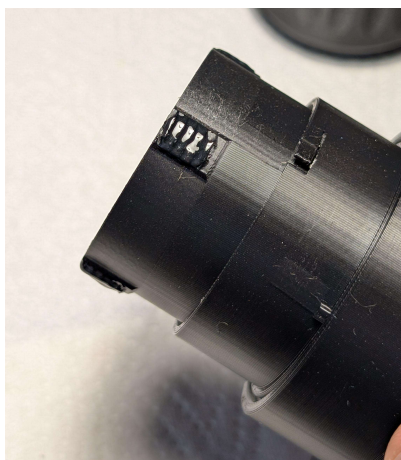
Insert second lens, pay attention to the lens orientation:



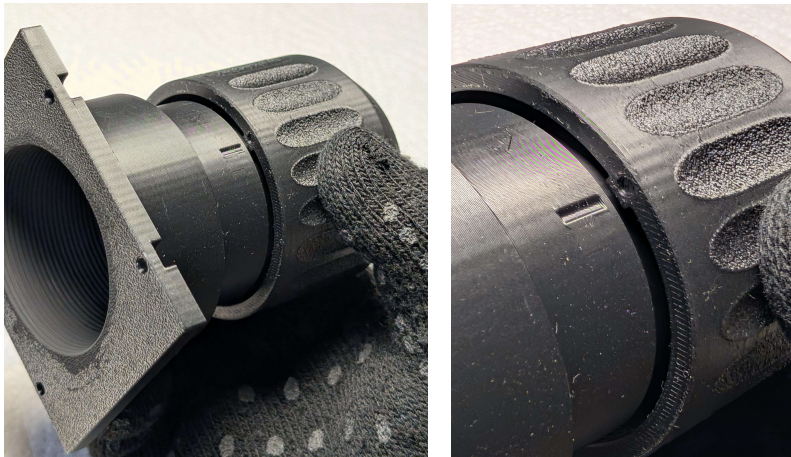
Press in fixation ring, pay attention to the ring orientation:



Place silicone grease on the thread carefully:



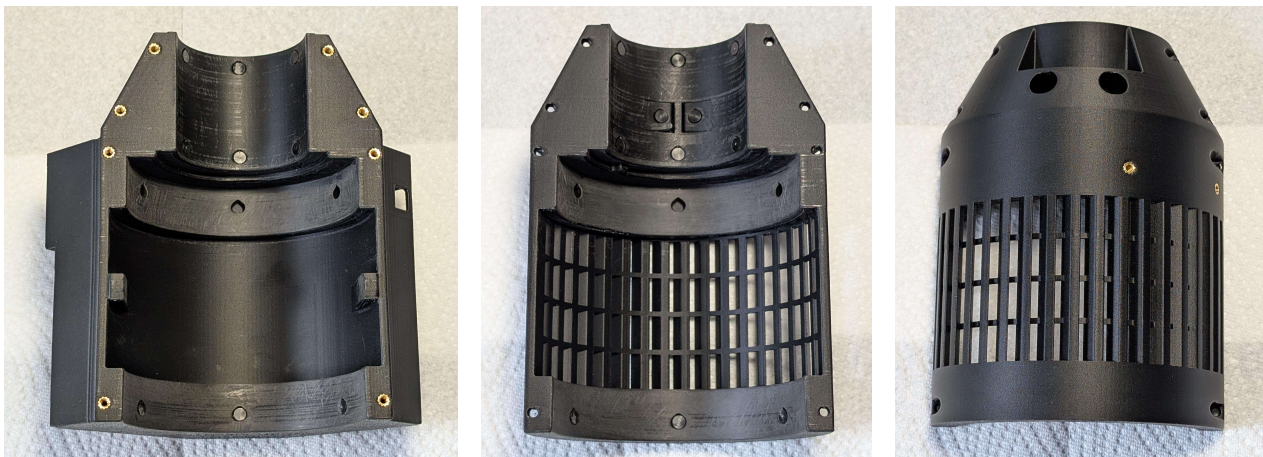
Screw on the lens holder, pay attention to the marks on both parts:



Assemble optical train:



Hot press in the M3 inserts. Here also the optional sliding inserts are used. These are turned POM cylinders roughly $\text{Ø}4.9 \times 4.7$ mm (exact dimensions measured on printed part). The assembly should work also without these sliding inserts.

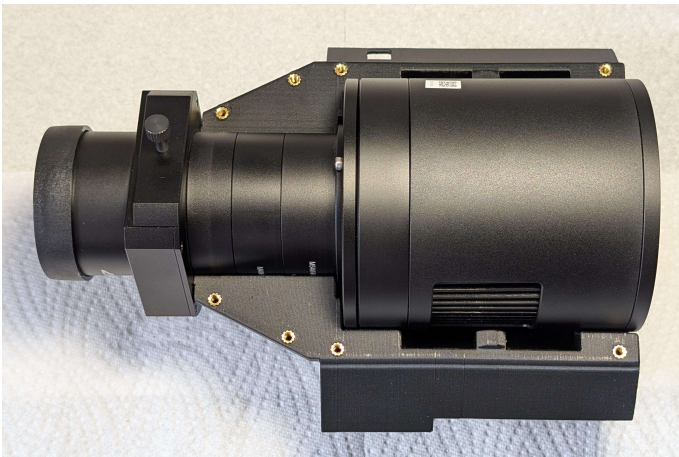


Other holes in plastic (M3 and M2.5) are meant to be tapped with corresponding threads. However, the geometry of holes allows to simply force screws in – less convenient and safe, but should also work.

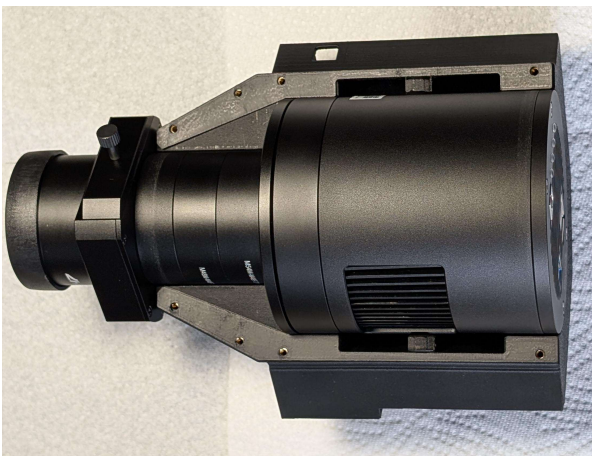
This screw (from package content of the camera) is used as a rotational stop:



Place assembled optical train in the half of outer housing:



Here additional pad spacers between housing halves are used to adjust free rotation of the housing around the camera:



Close the second half of the housing and fixate with M3x10 screws. Check for free rotation:

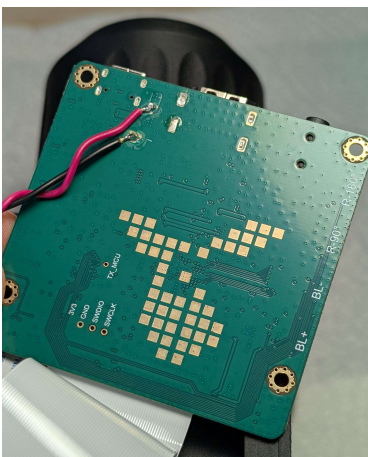


Place the springs over friction pads, and, together with stoppers, press them in, then fixate with M3x16 screws. BE CAREFUL!

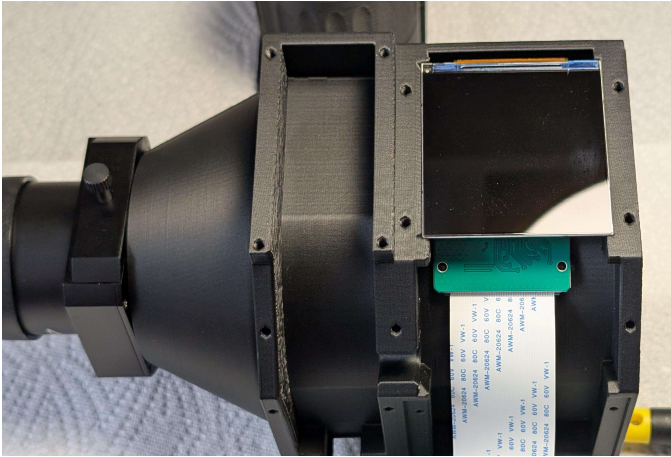
Depending on personal preference (how much friction you need during rotation) you can change the springs or e.g. install two springs with different diameter for each friction pad.



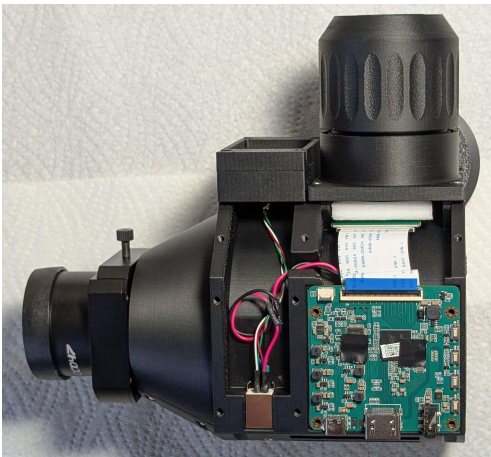
Solder power wires to the display HDMI board. These cables are soldered to USB-B receptacle connector together with short piece of microUSB cable in parallel (pinning according to USB standard):



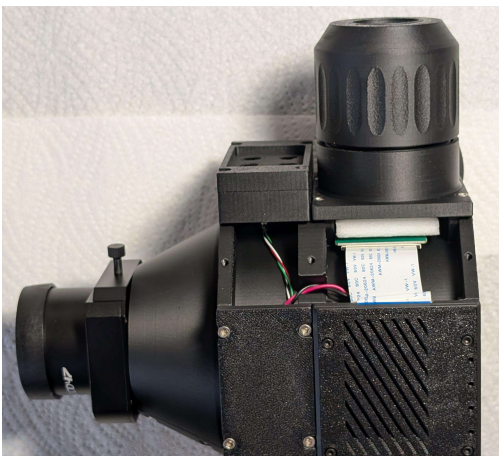
Place display screen with flat cable underneath (folded), and fixate the lens assembly over it with M3x10 screws:



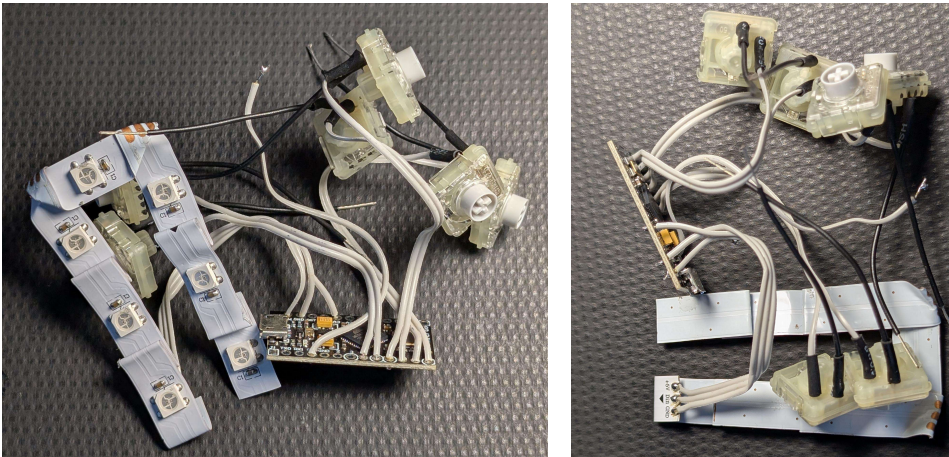
Place a piece of foam under the screen to isolate it from the PCB on the flat cable. Place flat cable under the HDMI controller PCB (folded):



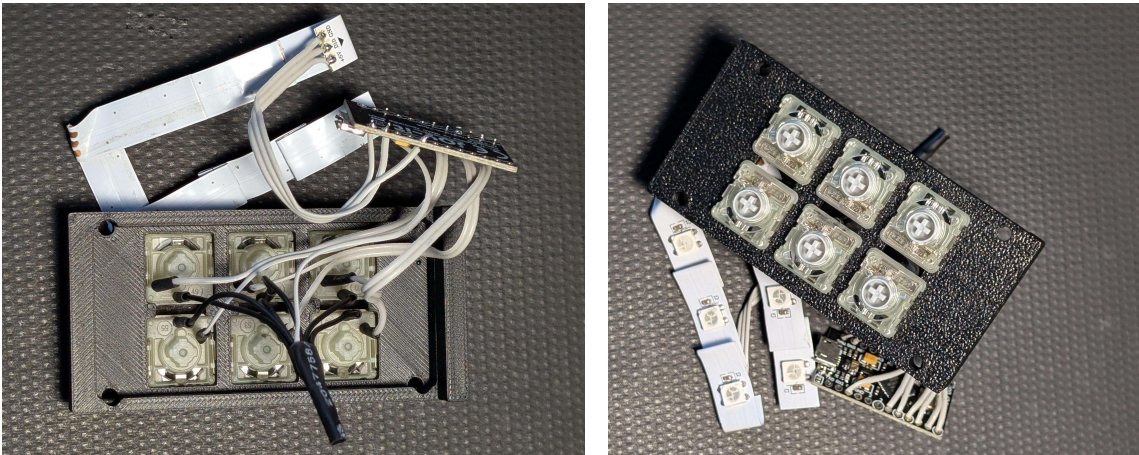
Fixate the HDMI controller PCB with M2.5x12 screws, and USB-B connector with cover panel and M3x10 screws:



Connect button switches and LED strip to Arduino PCB with wires. Pinning see in the source code file:



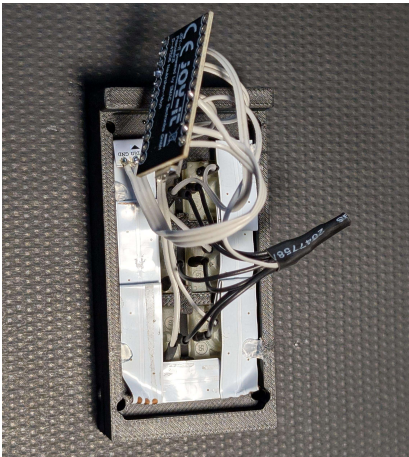
Place button switches in the bottom half of keyboard assembly and finalize soldering (common GND wire):



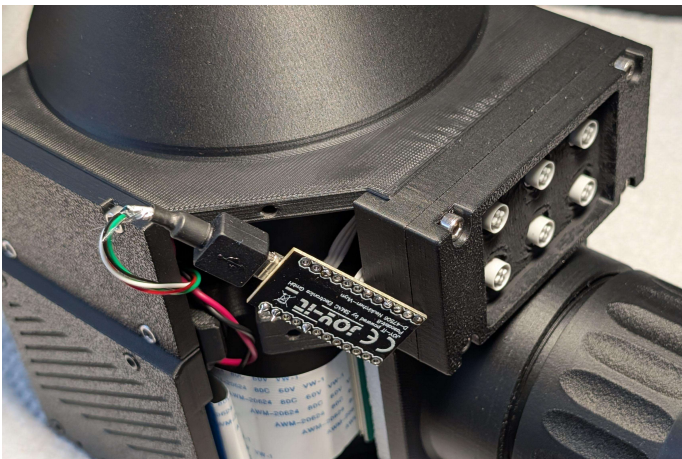
Fixate button switches with upper half of keyboard assembly:



Fixate LED strip with hot glue or similar. Pay attention to screw holes:



Fixate keyboard assembly on the housing with M3x16 screws, connect microUSB cable to Arduino:



Close the last housing cover with M3x10 screws:



Install (press in) the key caps:



Assemble cable holder on the side of the housing with M3x8 screws:



If you don't want to use keyboard and USB-B connector, device can be assembled without them using two alternative covers. In this case power for display is connected over USB-C connector of display controller.